The ENCODE Portal

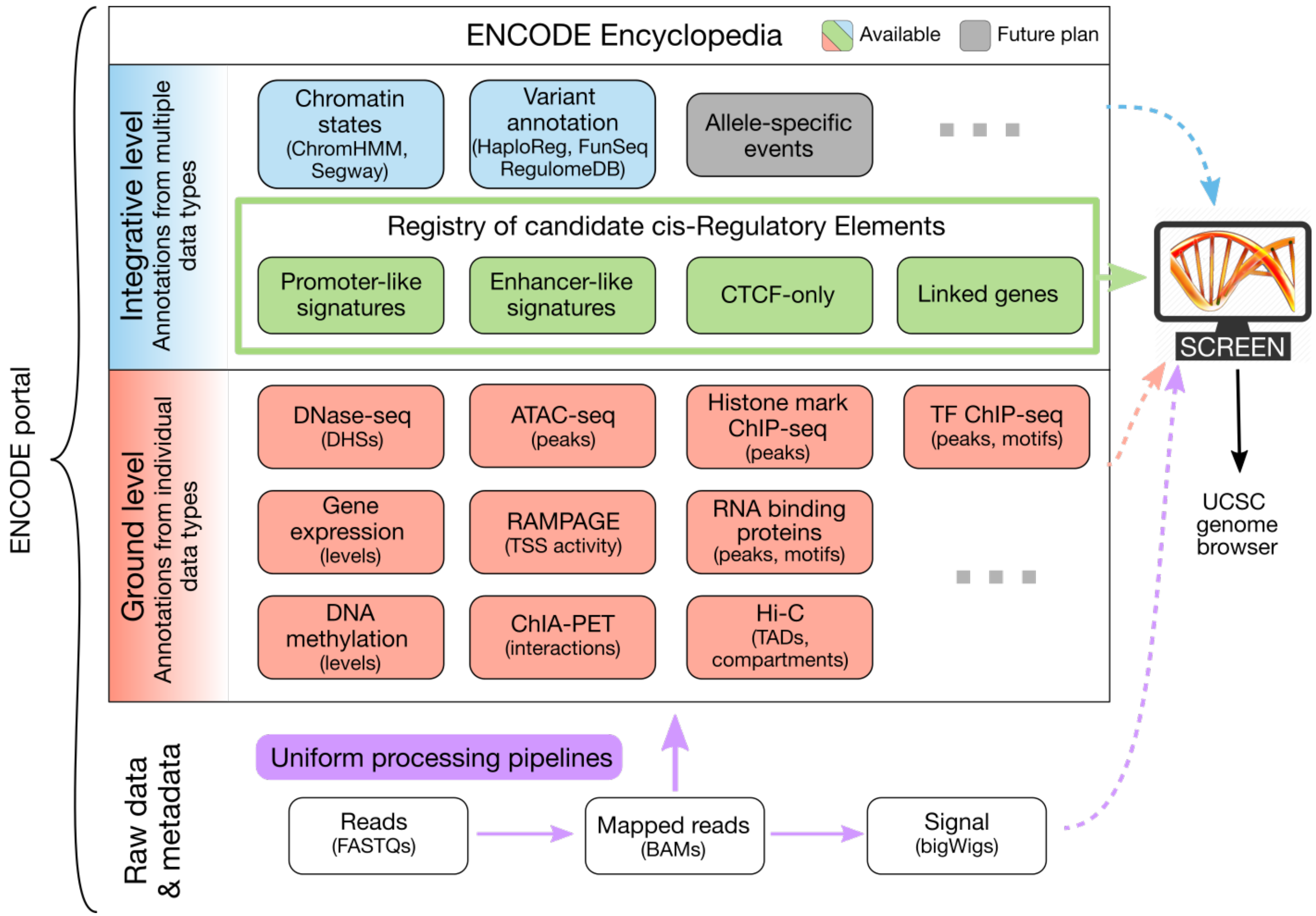
The ENCODE Portal

- The ENCODE Portal, developed and maintained by the Data Coordination Center (**ENCODE DCC**), is the canonical **source for all experimental metadata and data** from ENCODE and associated projects.
- The ENCODE Portal contains **raw and ground-level analysis data** generated by participating mapping centers using a wide-range of assays (integrative analysis data are available at the SCREEN portal).
- The Portal also **stores records of the materials and methods** used to perform the assays and subsequent analysis.
- **No account is needed** to view or download released data.

Portal quick links

- The ENCODE Portal contains the following types of data generated by the ENCODE Consortium:
  1. Data from a **wide-range of assays**, as well as protocol documents for each experiment
  2. **Antibody characterizations** that are performed as part of the antibody characterization process defined by the consortium
  3. **Reference filesets** used in the ENCODE uniform processing pipelines

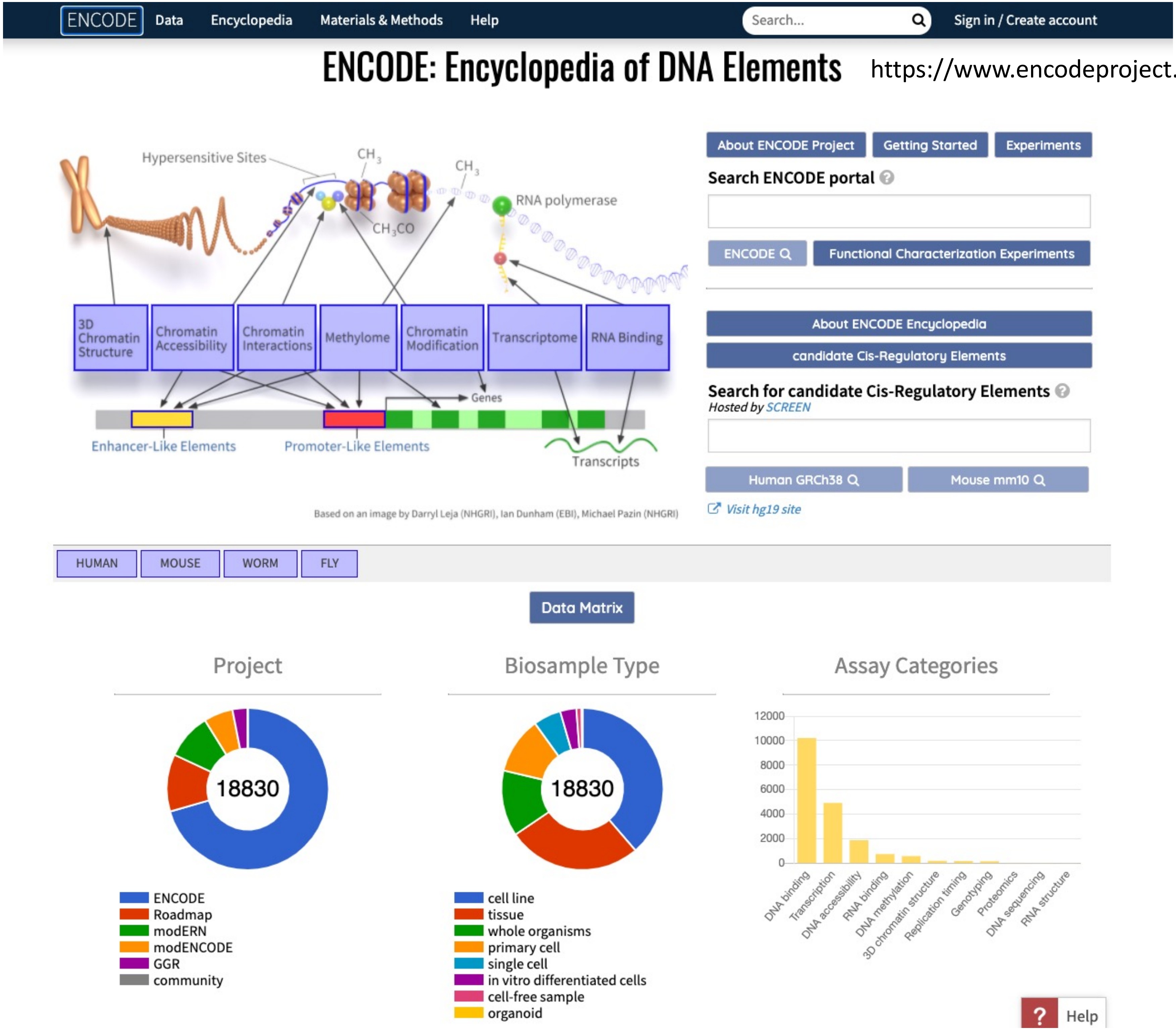
ENCODE Encyclopedia Version 5: Genomic and Transcriptomic Annotations

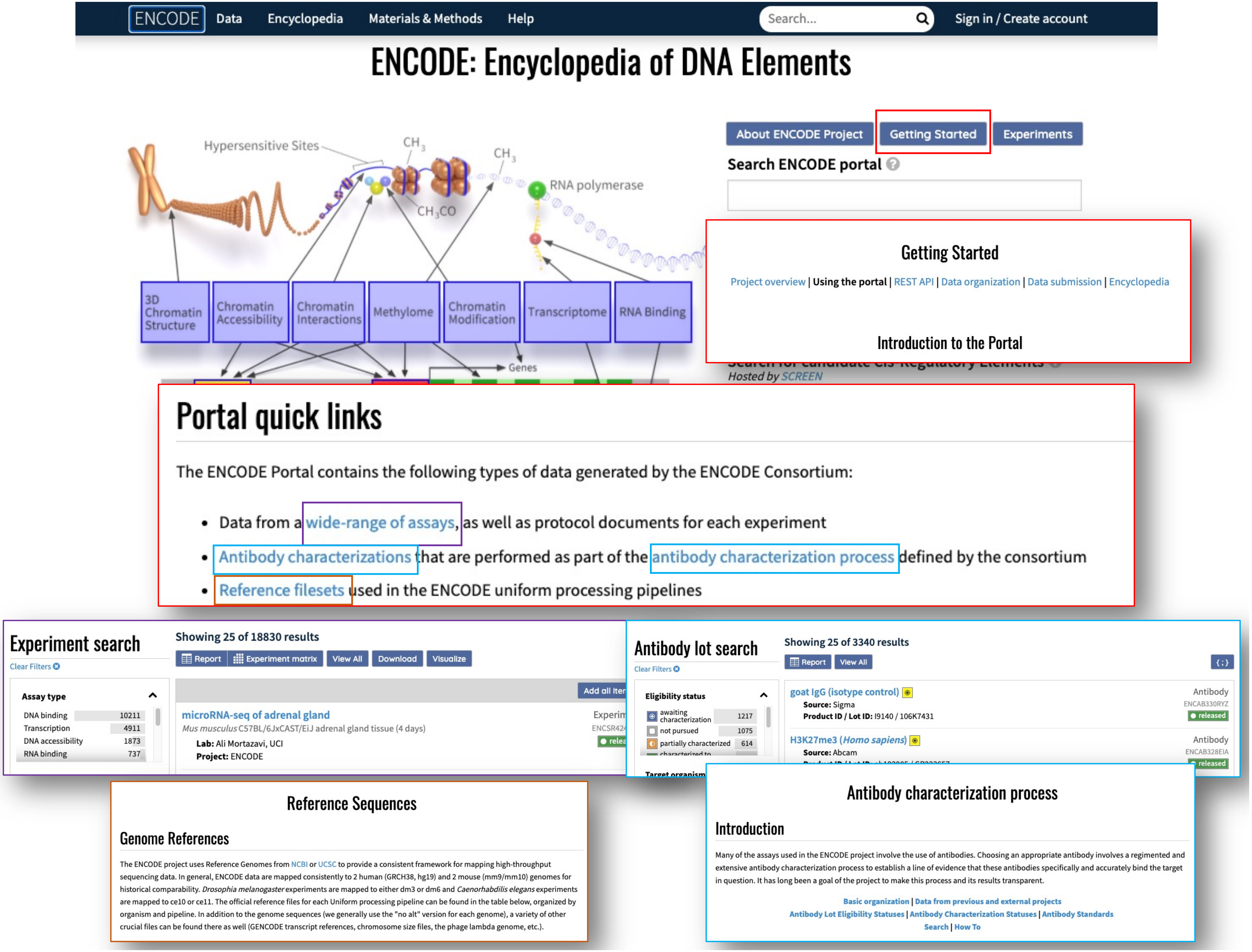


- The ENCODE Consortium not only **produces high-quality data**, but also **analyzes the data** in an integrative fashion. The ENCODE Encyclopedia organizes the most salient analysis products into annotations and **provides tools to search and visualize them**. The Encyclopedia has two levels of annotations:
- **Integrative-level annotations** integrate multiple types of experimental data and ground level annotations.
- **Ground-level annotations** are derived directly from the experimental data, typically produced by uniform processing pipelines.

ENCODE: Encyclopedia of DNA Elements

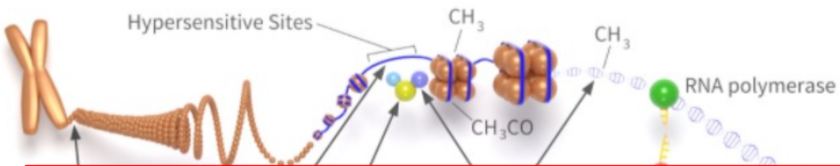
https://www.encodeproject.org/




ENCODE Data Encyclopedia Materials & Methods Help
Search...
Sign in / Create account
ENCODE: Encyclopedia of DNA Elements
Hypersensitive Sites
CH3
CH3CO
RNA polymerase
3D Chromatin Structure
Chromatin Accessibility
Chromatin Interactions
Methylome
Chromatin Modification
Transcriptome
RNA Binding
Genes
About ENCODE Project
Getting Started
Experiments
Search ENCODE portal
Getting Started
Project overview | Using the portal | REST API | Data organization | Data submission | Encyclopedia
Introduction to the Portal
Hosted by SCREEN
Portal quick links
The ENCODE Portal contains the following types of data generated by the ENCODE Consortium:
• Data from a wide-range of assays, as well as protocol documents for each experiment
• Antibody characterizations that are performed as part of the antibody characterization process defined by the consortium
• Reference filesets used in the ENCODE uniform processing pipelines
Experiment search
Clear Filters
Showing 25 of 18830 results
Report Experiment matrix View All Download Visualize
Assay type
DNA binding 10211
Transcription 4911
DNA accessibility 1873
RNA binding 737
microRNA-seq of adrenal gland
Mus musculus C57BL/6JxCAST/EiJ adrenal gland tissue (4 days)
Lab: Ali Mortazavi, UCI
Project: ENCODE
Antibody lot search
Clear Filters
Showing 25 of 3340 results
Report View All
Eligibility status
awaiting characterization 1217
not pursued 1075
partially characterized 614
goat IgG (isotype control)
Source: Sigma
Product ID / Lot ID: I9140 / 106K7431
Antibody
ENCAB330RYZ
released
H3K27me3 (Homo sapiens)
Source: Abcam
Antibody
ENCAB328EIA
released
Reference Sequences
Genome References
The ENCODE project uses Reference Genomes from NCBI or UCSC to provide a consistent framework for mapping high-throughput sequencing data. In general, ENCODE data are mapped consistently to 2 human (GRCH38, hg19) and 2 mouse (mm9/mm10) genomes for historical comparability. Drosophia melanogaster experiments are mapped to either dm3 or dm6 and Caenorhabdilis elegans experiments are mapped to ce10 or ce11. The official reference files for each Uniform processing pipeline can be found in the table below, organized by organism and pipeline. In addition to the genome sequences (we generally use the "no alt" version for each genome), a variety of other crucial files can be found there as well (GENCODE transcript references, chromosome size files, the phage lambda genome, etc.).
Antibody characterization process
Introduction
Many of the assays used in the ENCODE project involve the use of antibodies. Choosing an appropriate antibody involves a regimented and extensive antibody characterization process to establish a line of evidence that these antibodies specifically and accurately bind the target in question. It has long been a goal of the project to make this process and its results transparent.
Basic organization | Data from previous and external projects
Antibody Lot Eligibility Statuses | Antibody Characterization Statuses | Antibody Standards
Search | How To

ENCODE Data Encyclopedia Materials & Methods Help Search... Sign in / Create account

ENCODE: Encyclopedia of DNA Elements



About ENCODE Project | Getting Started | Experiments

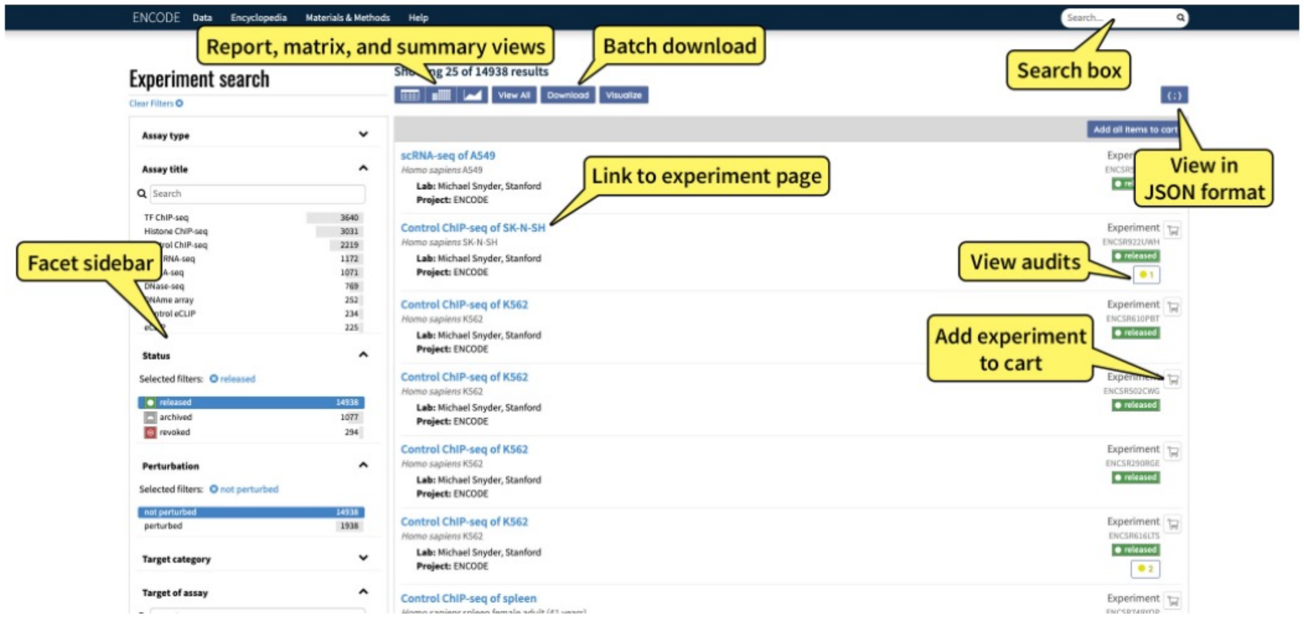
Search ENCODE portal

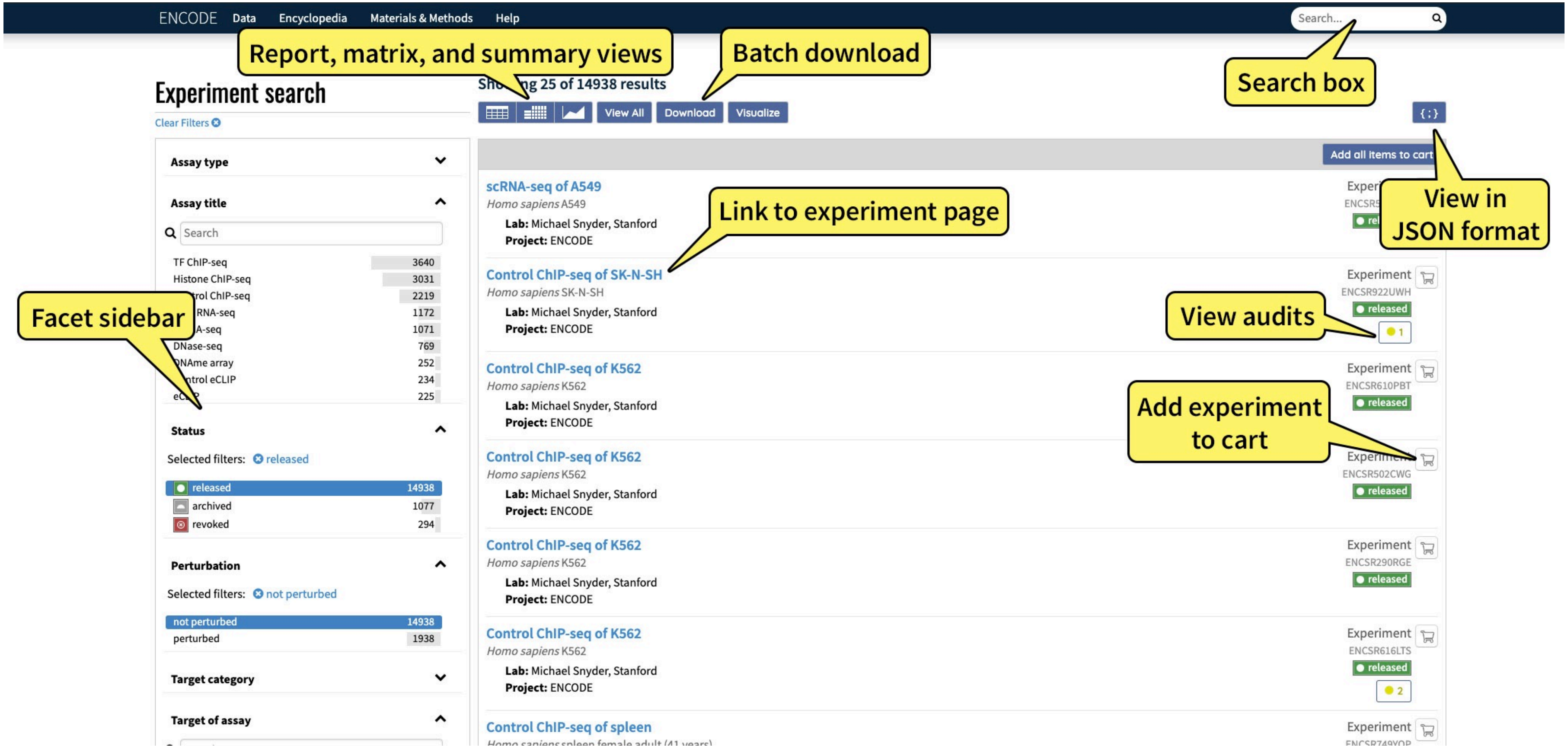
Using the Portal

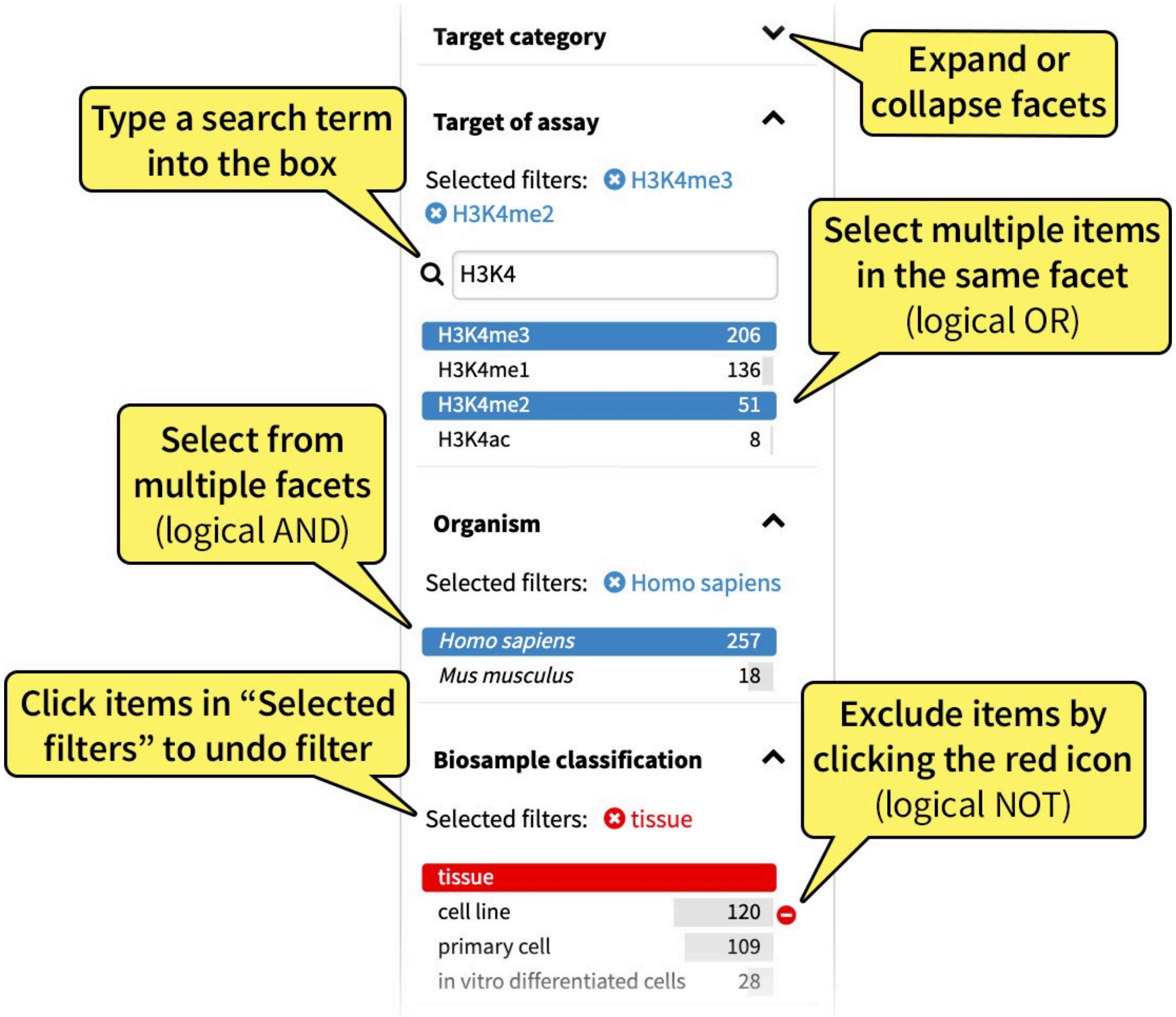
Browse for data | Visualize data | Download files

Browse and filter experiments

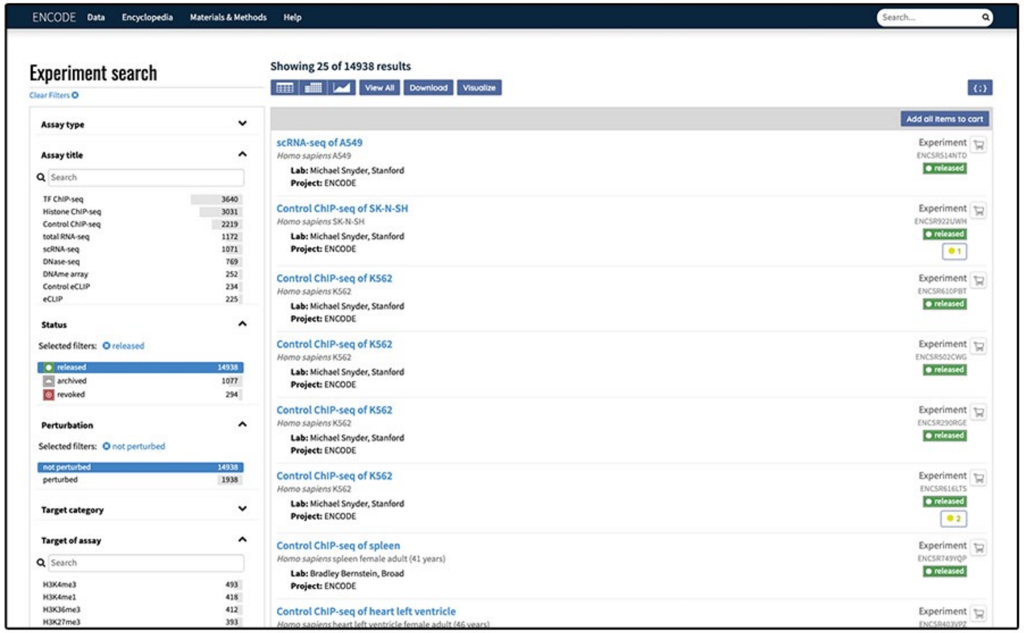
Clicking the "Search" option located in the "Data" toolbar menu located in the upper left corner brings up a list of all available experiments that have been used to generate ENCODE data. By default, search results are pre-filtered to experiments of status "released" (notice that in Figure 1, the facet term "released" is already highlighted). However, "archived" and "revoked" experiments are also publicly viewable. Explore the status terminology page for more information on what each status means.



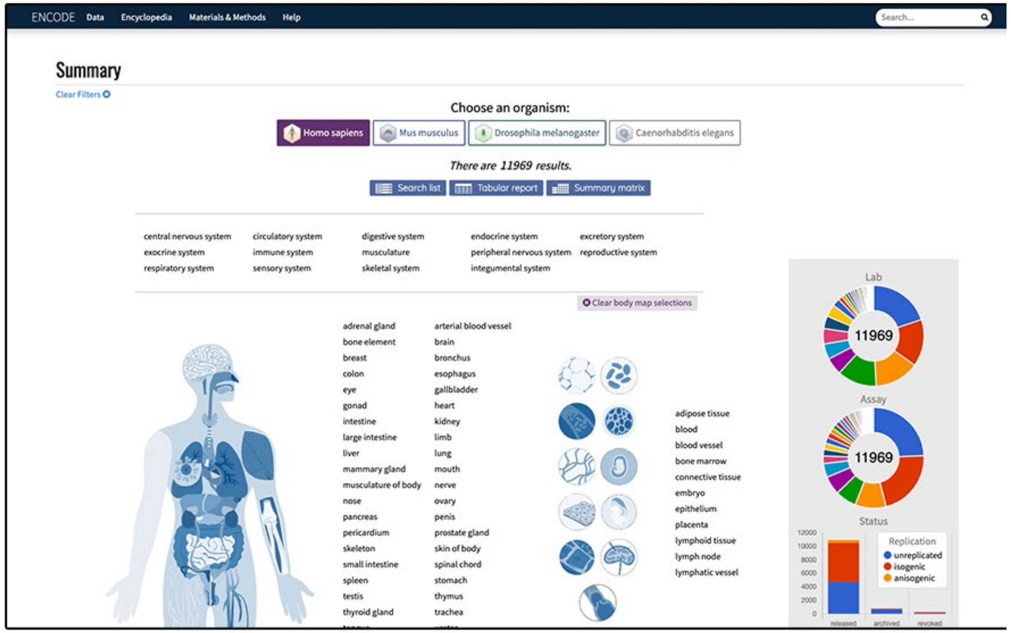

ENCODE
Data
Encyclopedia
Materials & Methods
Help
Search...
Report, matrix, and summary views
Batch download
Search box
Experiment search
Clear Filters
Showing 25 of 14938 results
View All
Download
Visualize
Add all items to cart
View in JSON format
Link to experiment page
Facet sidebar
View audits
Add experiment to cart
Assay type
Assay title
Search
TF ChIP-seq 3640
Histone ChIP-seq 3031
2219
1172
1071
DNase-seq 769
252
234
225
Status
Selected filters: released
released 14938
archived 1077
revoked 294
Perturbation
Selected filters: not perturbed
not perturbed 14938
perturbed 1938
Target category
Target of assay
scRNA-seq of A549
Homo sapiens A549
Lab: Michael Snyder, Stanford
Project: ENCODE
Control ChIP-seq of SK-N-SH
Homo sapiens SK-N-SH
Lab: Michael Snyder, Stanford
Project: ENCODE
Experiment
ENCSR922UWH
released
Control ChIP-seq of K562
Homo sapiens K562
Lab: Michael Snyder, Stanford
Project: ENCODE
Experiment
ENCSR610PBT
released
Control ChIP-seq of K562
Homo sapiens K562
Lab: Michael Snyder, Stanford
Project: ENCODE
ENCSR502CWG
released
Control ChIP-seq of K562
Homo sapiens K562
Lab: Michael Snyder, Stanford
Project: ENCODE
Experiment
ENCSR290RGE
released
Control ChIP-seq of K562
Homo sapiens K562
Lab: Michael Snyder, Stanford
Project: ENCODE
Experiment
ENCSR616LTS
released
Control ChIP-seq of spleen
Experiment


Target category
Expand or collapse facets
Target of assay
Type a search term into the box
Selected filters: H3K4me3 H3K4me2
H3K4
Select multiple items in the same facet (logical OR)
H3K4me3 206
H3K4me1 136
H3K4me2 51
H3K4ac 8
Select from multiple facets (logical AND)
Organism
Selected filters: Homo sapiens
Homo sapiens 257
Mus musculus 18
Click items in "Selected filters" to undo filter
Biosample classification
Exclude items by clicking the red icon (logical NOT)
Selected filters: tissue
tissue
cell line 120
primary cell 109
in vitro differentiated cells 28

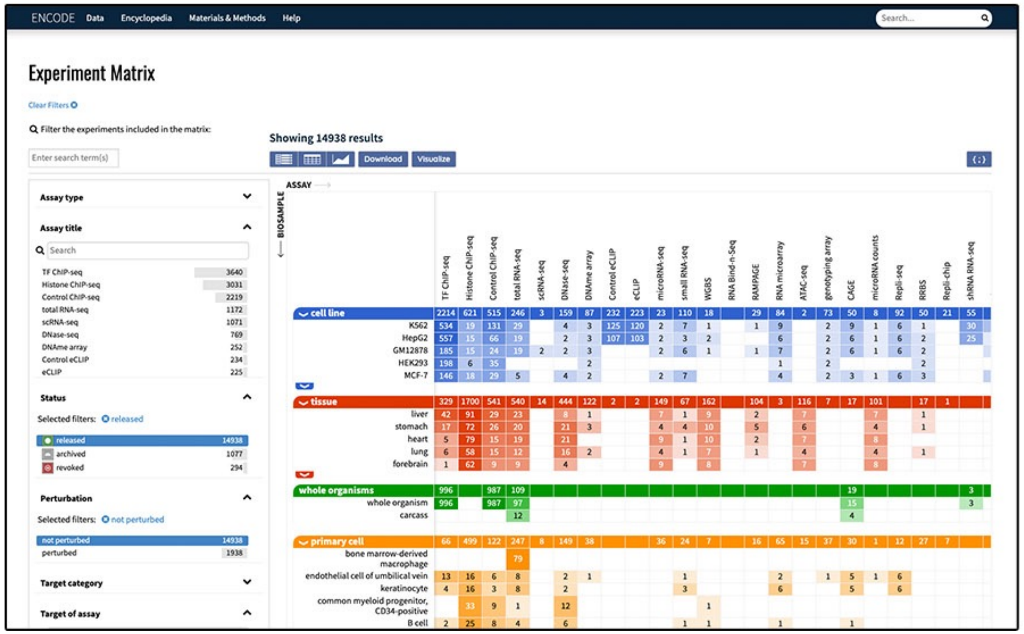
List view



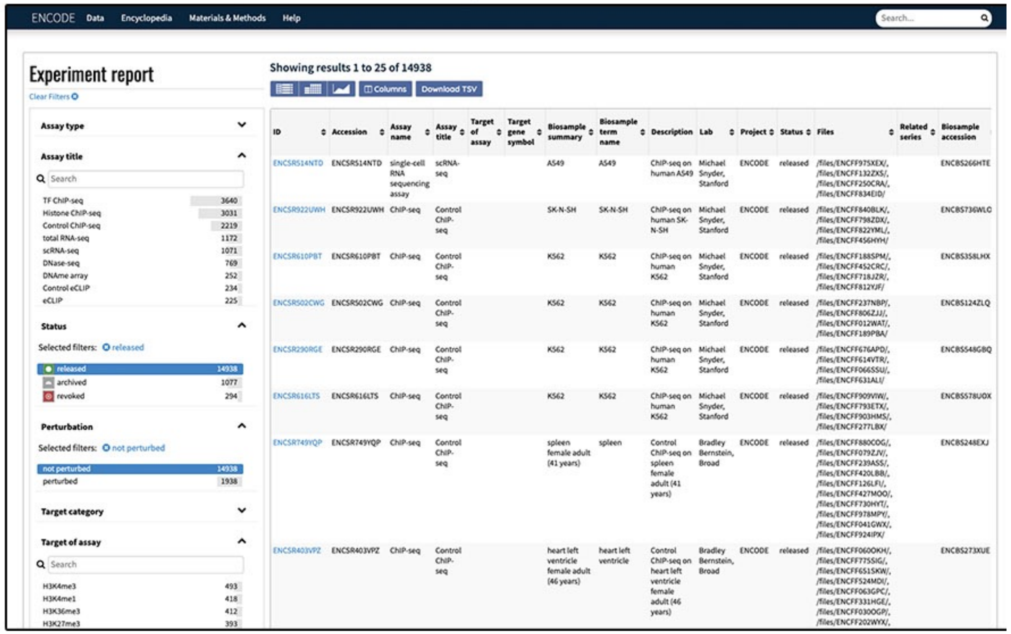
Summary view



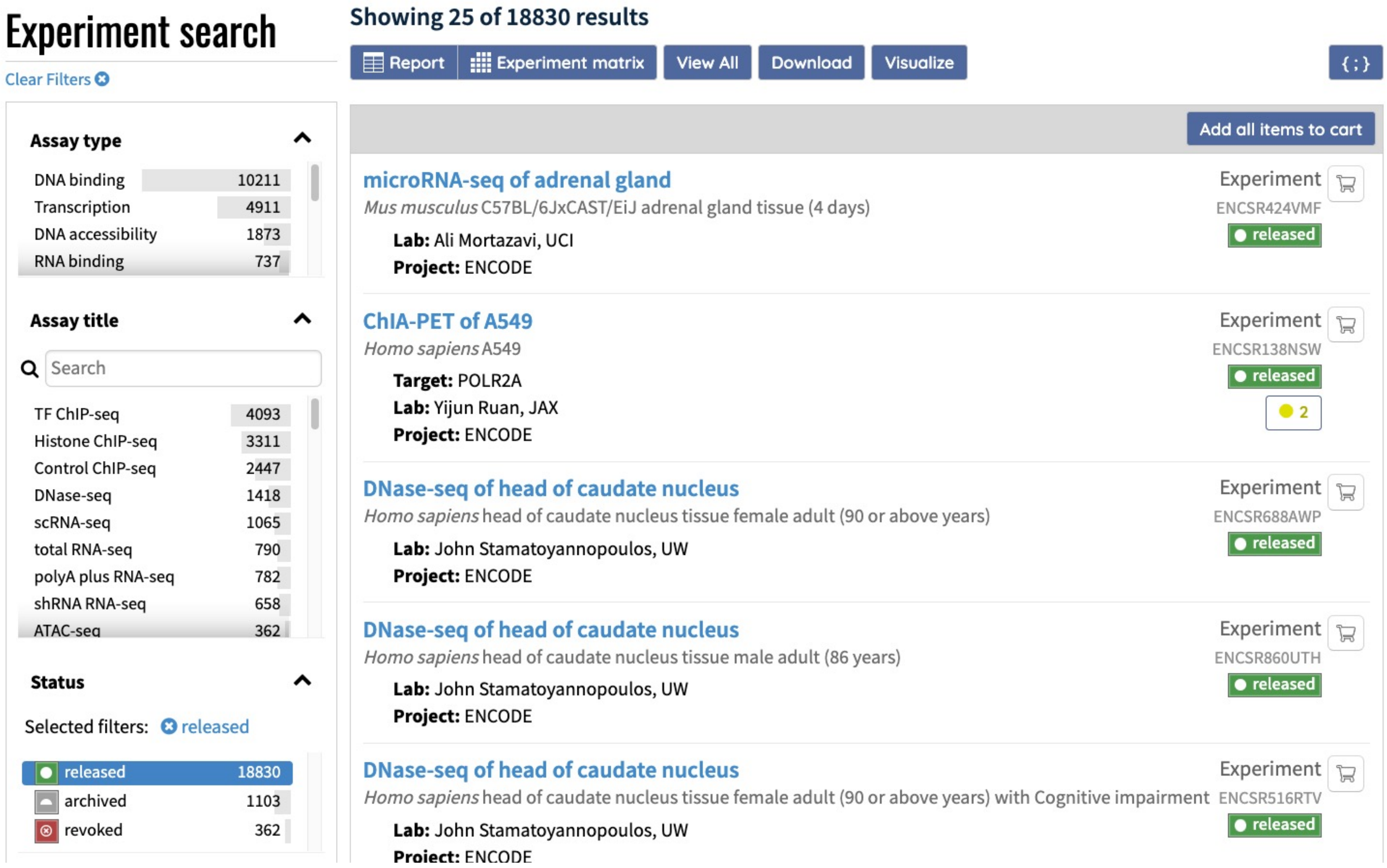
Matrix view



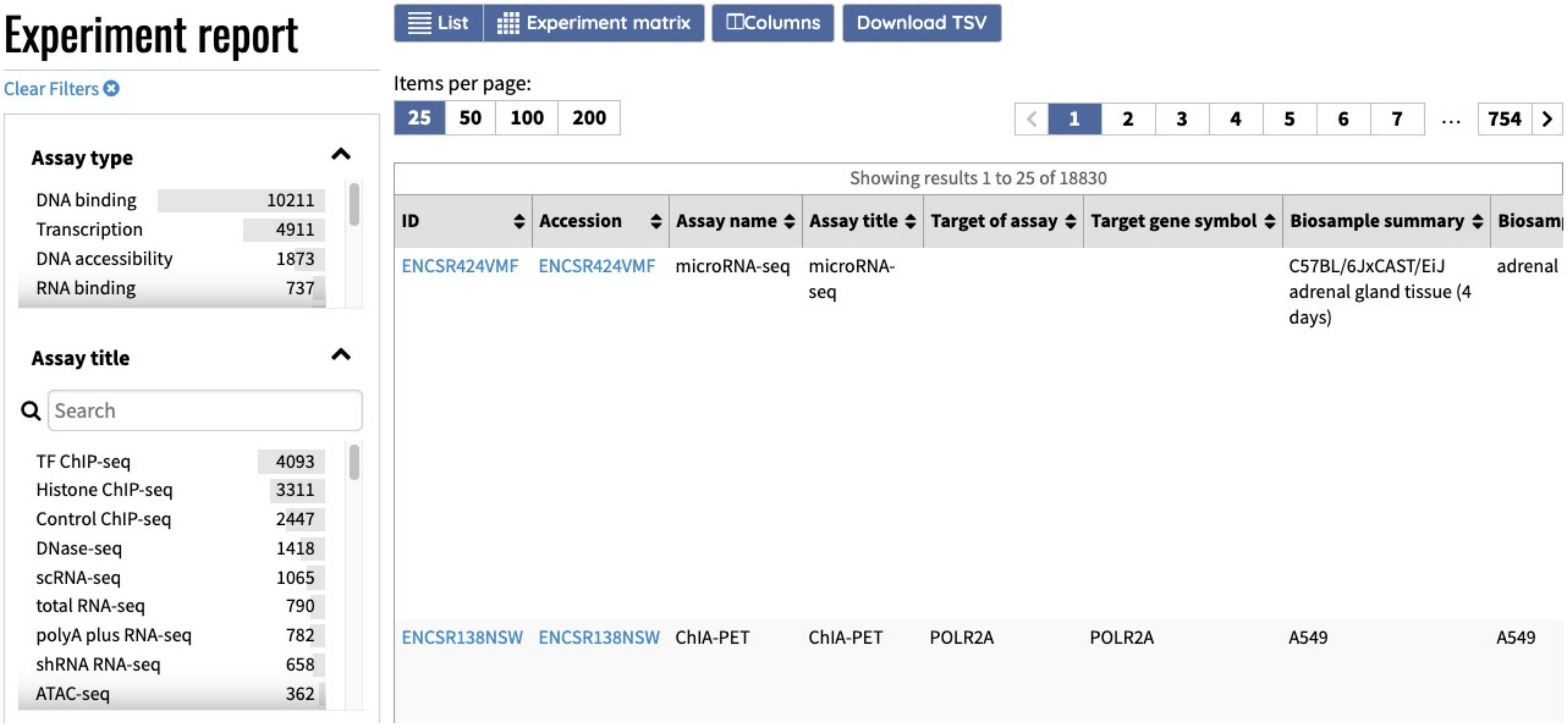
Report view



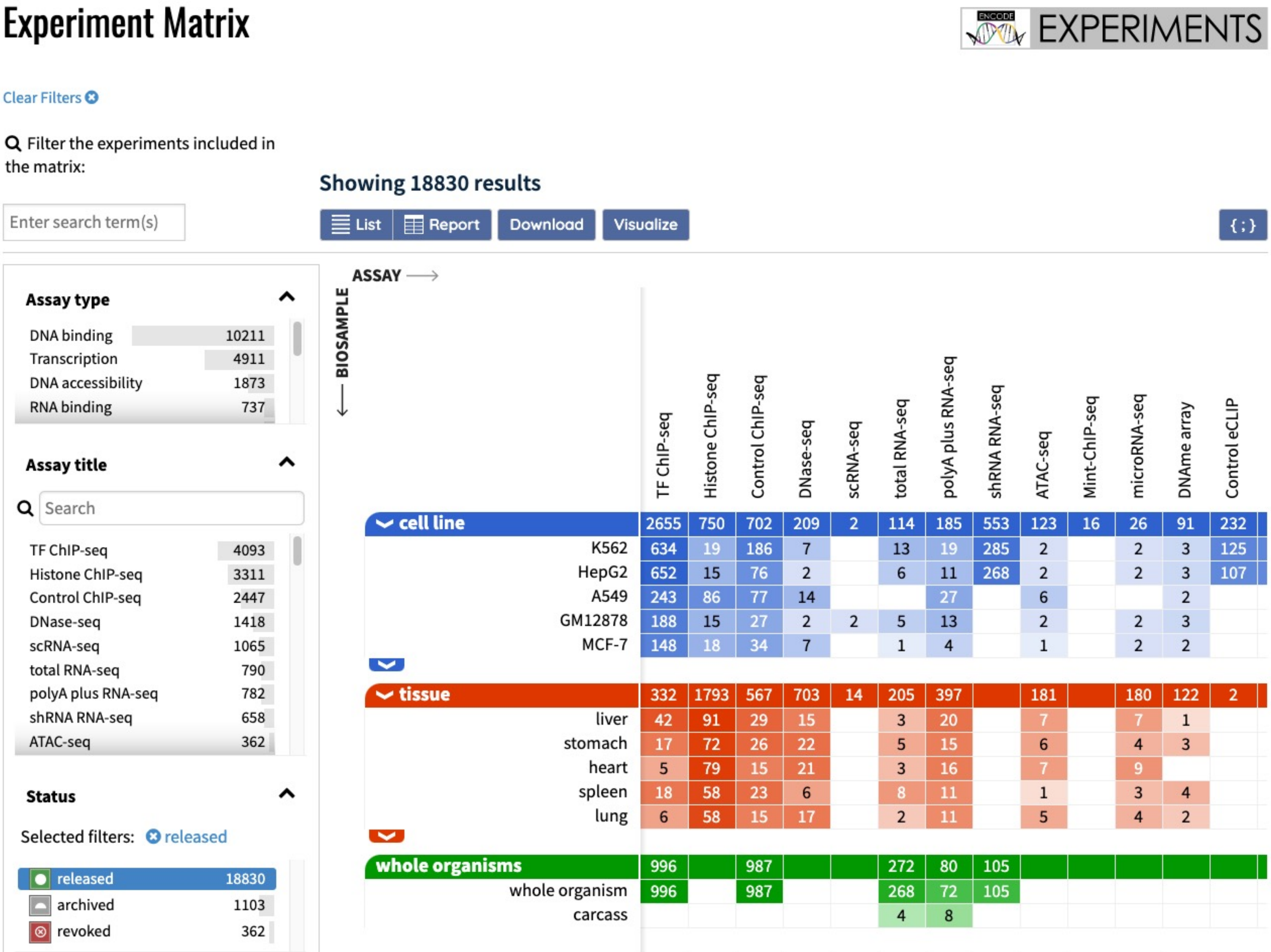
• List view: displays results in a list. Each experiment is labeled with a summary of assay and biosample used.



•Report view: displays results in tabular format, with a default selection of metadata properties as the columns.

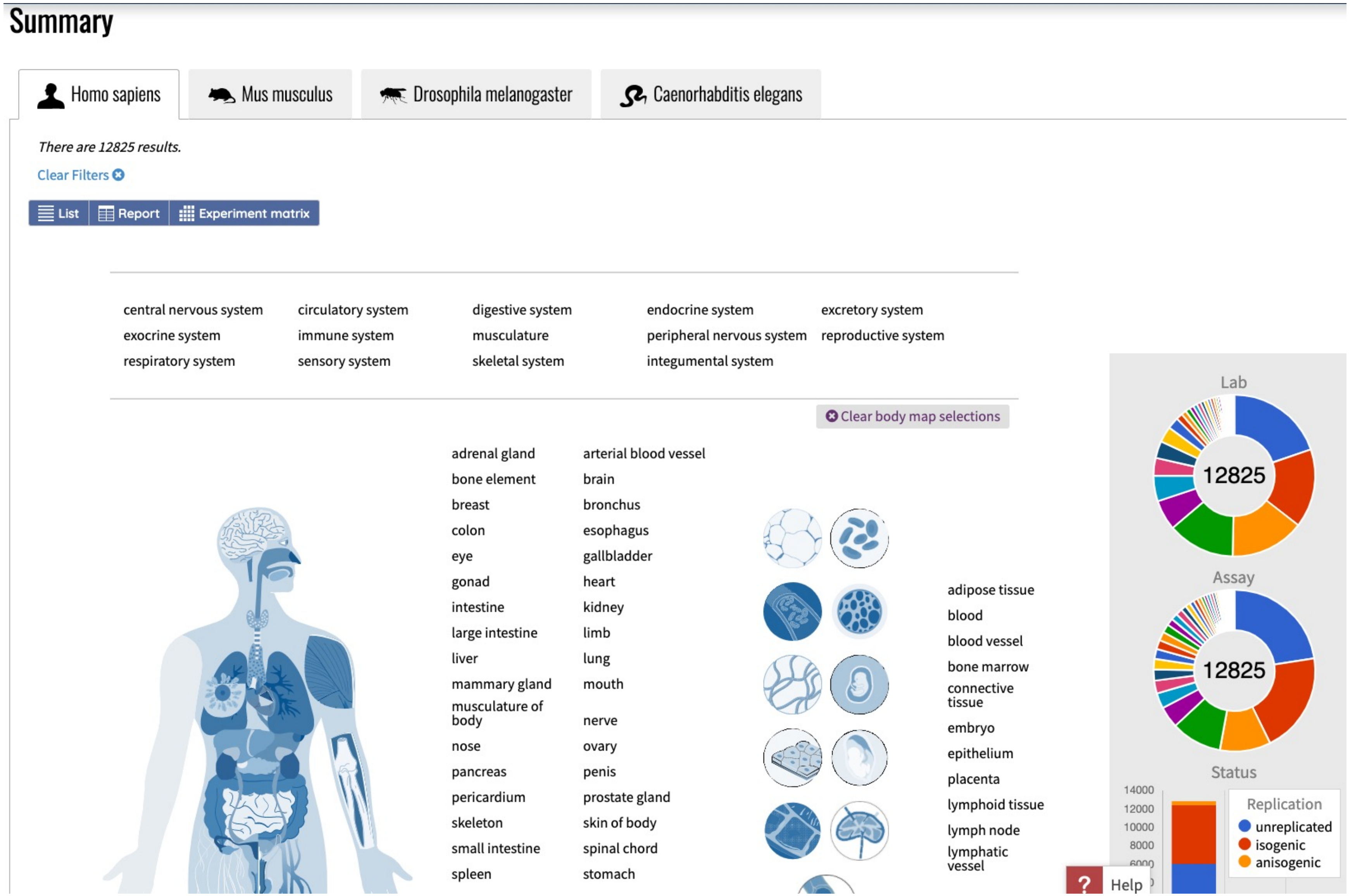


•Matrix view: displays results in a matrix, organized with biosamples along the y-axis and assay type along the x-axis.



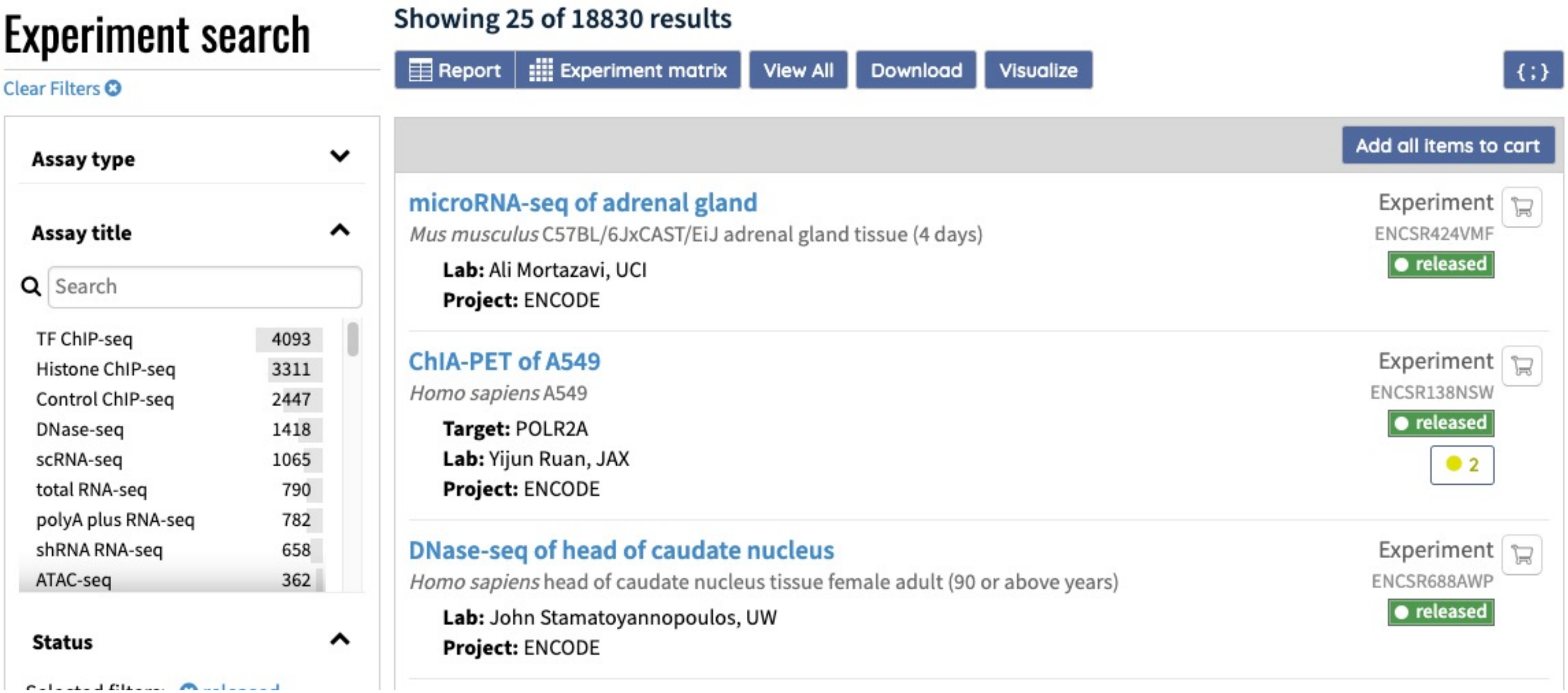
BIOSAMPLE	TF ChIP-seq	Histone ChIP-seq	Control ChIP-seq	DNase-seq	scRNA-seq	total RNA-seq	polyA plus RNA-seq	shRNA RNA-seq	ATAC-seq	Mint-ChIP-seq	microRNA-seq	DNAme array	Control eCLIP
cell line	2655	750	702	209	2	114	185	553	123	16	26	91	232
K562	634	19	186	7		13	19	285	2		2	3	125
HepG2	652	15	76	2		6	11	268	2		2	3	107
A549	243	86	77	14			27		6			2	
GM12878	188	15	27	2	2	5	13		2		2	3	
MCF-7	148	18	34	7		1	4		1		2	2	
tissue	332	1793	567	703	14	205	397		181		180	122	2
liver	42	91	29	15		3	20		7		7	1	
stomach	17	72	26	22		5	15		6		4	3	
heart	5	79	15	21		3	16		7		9		
spleen	18	58	23	6		8	11		1		3	4	
lung	6	58	15	17		2	11		5		4	2	
whole organisms	996		987			272	80	105					
whole organism	996		987			268	72	105					
carcass						4	8						

- Summary view: displays a general overview of the data in chart form. A body map diagram is also available for human data.



1. Data from a wide-range of assays

- https://www.encodeproject.org/search/?type=Experiment&status=released



1. Data from a wide-range of assays

Experiments / ChIA-PET / *Homo sapiens* / A549

Experiment summary for ENCSR138NSW

2

{;}

Summary	
Status:	released
Assay:	ChIA-PET
Target:	POLR2A
Biosample summary:	*Homo sapiens* A549
Biosample Type:	cell line
Replication type:	isogenic
Description:	RNAPII ChIA-PET in A549
Nucleic acid type:	DNA
Size range:	350-600
Extraction method:	SDS
Size selection method:	BluePippin
Platform:	Illumina NovaSeq 6000
Controls:	ENCSR414IDX

Attribution	
Lab:	Yijun Ruan, JAX
Award:	UM1HG009409 (Yijun Ruan, JAX)
Project:	ENCODE
Aliases:	yijun-ruan:A549_RNAPII
Date submitted:	January 12, 2021
Date released:	January 12, 2021

1. Data from a wide-range of assays

Isogenic replicates

Isogenic replicate	Technical replicate	Summary	Biosample	Antibody	Library
1	1	A549 cell line	ENCBS085MQM	ENCAB725WBH	ENCLB877LSQ
2	1	A549 cell line	ENCBS744PGM	ENCAB725WBH	ENCLB430DDY

ENCBS085MQM / cell line

{;}

Summary

Status:	released
Term name:	A549
Term ID:	EFO:0001086
Summary:	*Homo sapiens* A549 cell line
Description:	Human epithelial cell line derived from a lung carcinoma tissue. The cells are adherent in culture. The karyotype is hypotriploid male with a modal chromosome number of 12. There are numerous chromosome abnormalities and marker chromosomes.

Attribution

ENCODE PHASE 4

Lab:	Yijun Ruan, JAX
Award PI:	Yijun Ruan, JAX
Submitted by:	Daniel Capurso
Source:	ATCC
Project:	ENCODE
External resources:	Cellosaurus:CVCL_0023
Aliases:	yijun-ruan:A549_cell_culture_biorep1

Donor information

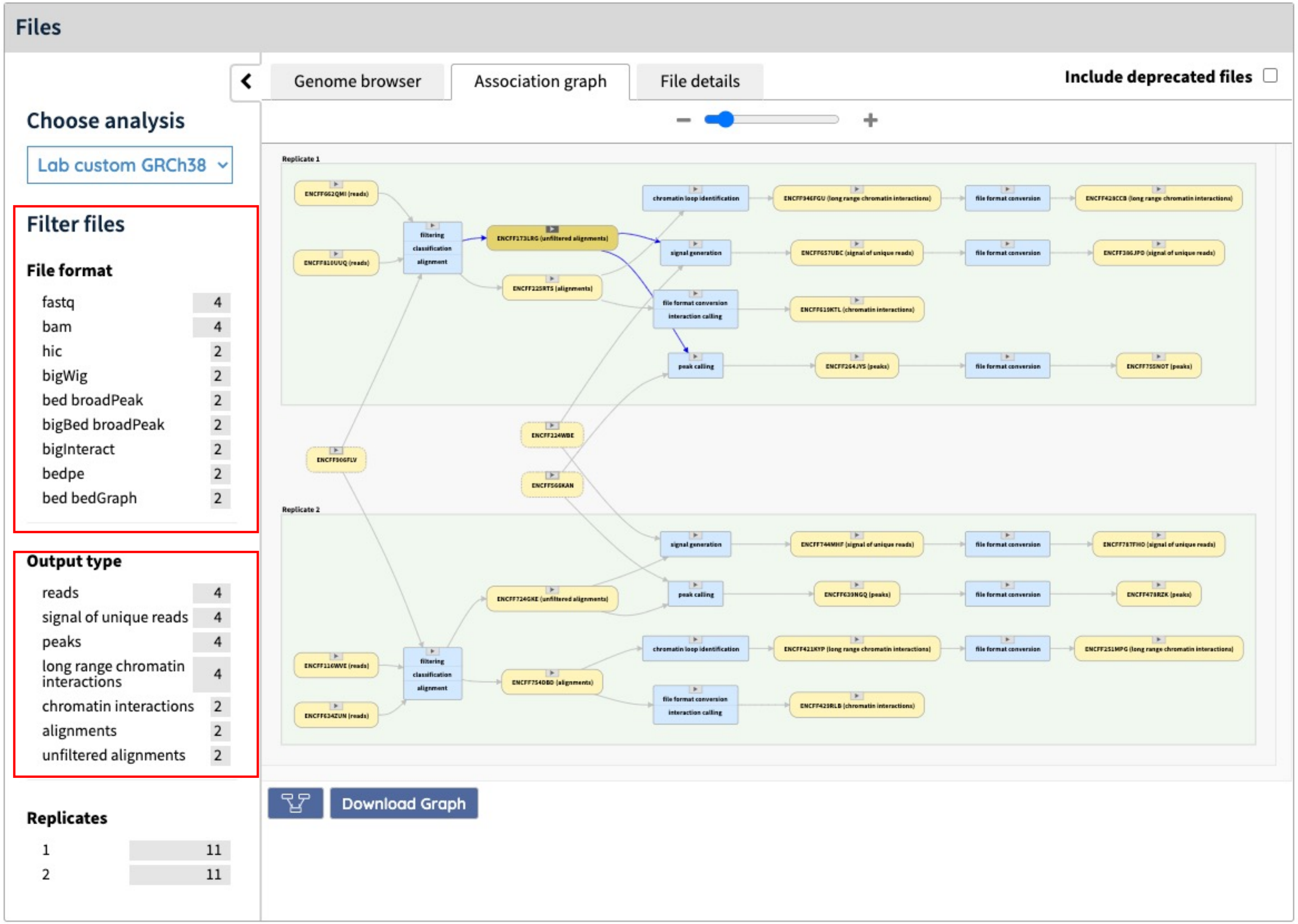
Status:	released
Accession:	ENCDO000AAZ
Aliases:	encode:donor of A549, bradley-bernstein:Donor of A549 cells, john-stamatoyannopoulos:ATCC_A549
Species:	*Homo sapiens*
Life stage:	Adult
Age:	58 years
Sex:	Male
Ethnicity:	Caucasian
External resources:	GEO:SAMN05733878
References:	PMID:4357758

Functional genomics experiments using this biosample

Accession	Assay	Biosample term name	Target	Description	Lab
ENCSR138NSW	ChIA-PET	A549	POLR2A	RNAPII ChIA-PET in A549	Yijun Ruan, JAX
ENCSR911ZMB	ChIA-PET	A549	CTCF	CTCF ChIA-PET in A549	Yijun Ruan, JAX

Displaying 2 of 2

1. Data from a wide-range of assays



1.Data from a wide-range of assays

Documents

Pipeline Protocol

Description: Ruan Lab ChIA-PIPE processing pipeline for ChIA-PET data

ENCODE_ChIA_PET_pipeline_readme_1217202

General Protocol

Description: Ruan Lab in-situ ChIA-PET protocol

In-situ chiapet_Ruan Lab_final.pdf

ENCODE, ChIA-PET
Data processing pipeline description for Ruan Lab

ChIA-PIPE is a fully automated data processing pipeline to process ChIA-PET data for both in-situ ChIA-PET and long-read ChIA-PET (Lee et al., 2020 Genome Biology. doi:10.1126/sciadv.aay2078). The complete code is available in github site (https://github.com/TheJacksonLaboratory/ChIA-PIPE).

1. Pipeline install and launch the process

After users download and install ChIA-PIPE from the github site, they can fill in pipeline parameters in a config file, and then launch ChIA-PIPE on HPC. Key pipeline parameters in a config file follow, using a GM10248 CTCF ChIA-PET experiment as an example (ENCODE accession: ENCSR627XQX):

```
# The name of the sequencing run
run="LHG0047H"

# The type of sequencing run:
#    "miseq" - around 30 million reads
#    "hiseq" - around 300 million reads
#    "pooled" - around 1 billion reads
# It adjusts a proper computing resource (processors, memory, and walltime).
run_type="hiseq"
```

In situ ChIA-PET protocol (Ruan Lab 18/06/2020)

Background

ChIA-PET is a proven method to map protein-mediated chromatin interactions. The initial ChIA-PET method was developed based ChIP-PET [Wei et al., 2006] for mapping transcription factor (TF) genome-wide occupancy by adding a proximity ligation to the ChIPed chromatin material, so to establish the connectivity between chromatin fragments tethered together by protein factors [Fullwood et a., 2009]. The original ChIA-PET protocol employed dual linkers in proximity ligation to distinguish non-specific ligation and a type IIS restriction enzyme digestion (MmeI) to generate a 20bp tag of chromatin DNA on each side of the dual-linker ligation products, and the paired-end-tag (PET, tag-linker-tag) templates were subjected for high throughput paired end sequencing and mapping to demarcated long-range chromatin interactions [Fullwood et a., 2010]. To increase the mapping PET mapping efficiency, we employed random digestion by Tn5 transpotase to the proximity ligation products to generate longer templates (tagmentation) for 2X250 bp PET sequencing [Li et al., 2017]. We have shown that long-read ChIA-PET significantly increased mapping efficiency and allow for detection of allele-specific chromatin interactions at single-base-pair resolution [Tang et al., 2015]. Inspired by in situ Hi-C protocol, we have now incorporated the steps of in situ digestion and in situ proximity ligation into the ChIA-PET protocol, so the in situ ChIA-PET, which showed significant reduction of non-specific proximity ligation and reduced the required cell numbers as low as 1 million cells per ChIA-PET experiment comparing to the previous ChIA-PET protocol. We have applied in situ ChIA-PET protocol in the ENCODE project and have to generated more than hundred ChIA-PET datasets associated with CTCF and RNAPII from ENCODE biosamples. Here, we describe the outline and the detailed step-by-step procedure of the in situ ChIA-PET protocol.

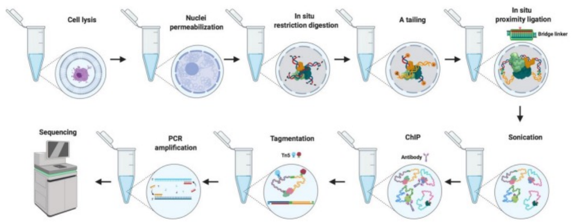
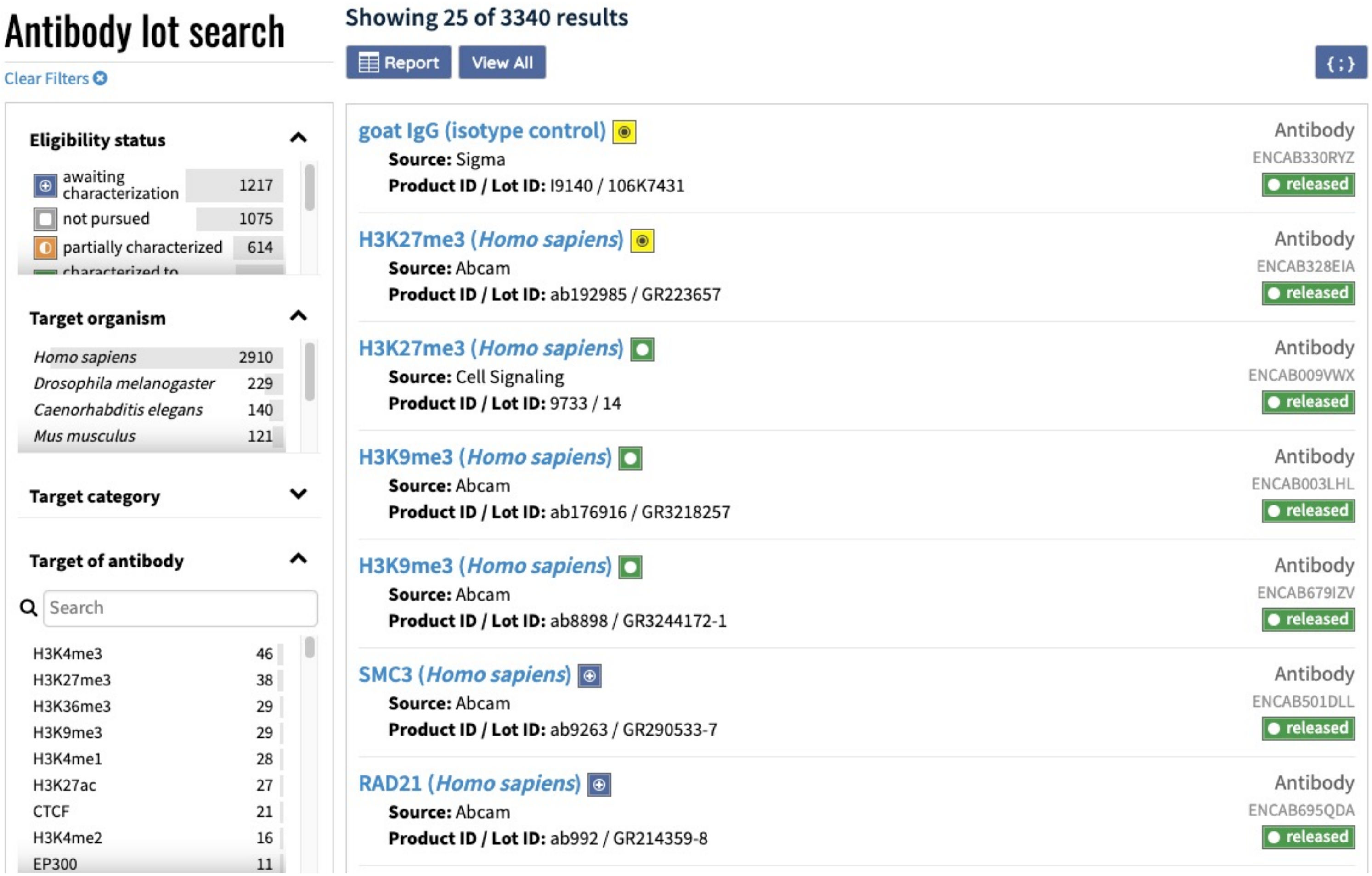


Figure 1. Schematic of in situ ChIA-PET.

2. Antibody characterizations

- https://www.encodeproject.org/search/?type=AntibodyLot&status=released



2. Antibody characterizations

- https://www.encodeproject.org/search/?type=AntibodyLot&status=released

Antibodies / *Homo sapiens* / HIST2H3A + HIST2H3D + HIST2H3C

ENCAB328EIA

Antibody against *Homo sapiens* H3K27me3

{;}

Homo sapiens any cell type or tissue

characterized to standards with exemption

Status: released

Source (vendor): Abcam

Product ID: ab192985

Lot ID: GR223657

Characterized targets: H3K27me3 (*Homo sapiens*)

Host: Rabbit

Clonality: Monoclonal

Aliases: bradley-bernstein:PchAb 1393

Functional genomics experiments using this antibody

Accession	Assay	Biosample term name	Target	Description	Lab
ENCSR024JJL	ChIP-seq	esophagus squamous epithelium	H3K27me3	H3K27me3 ChIP-seq on esophagus squamous epithelium male adult (54 years)	Bradley Bernstein, Broad

Displaying 1 of 1

Characterizations

H3K27me3 (*Homo Sapiens*)

Method: dot blot assay

exempt from standards

H3K27me3 (*Homo Sapiens*)

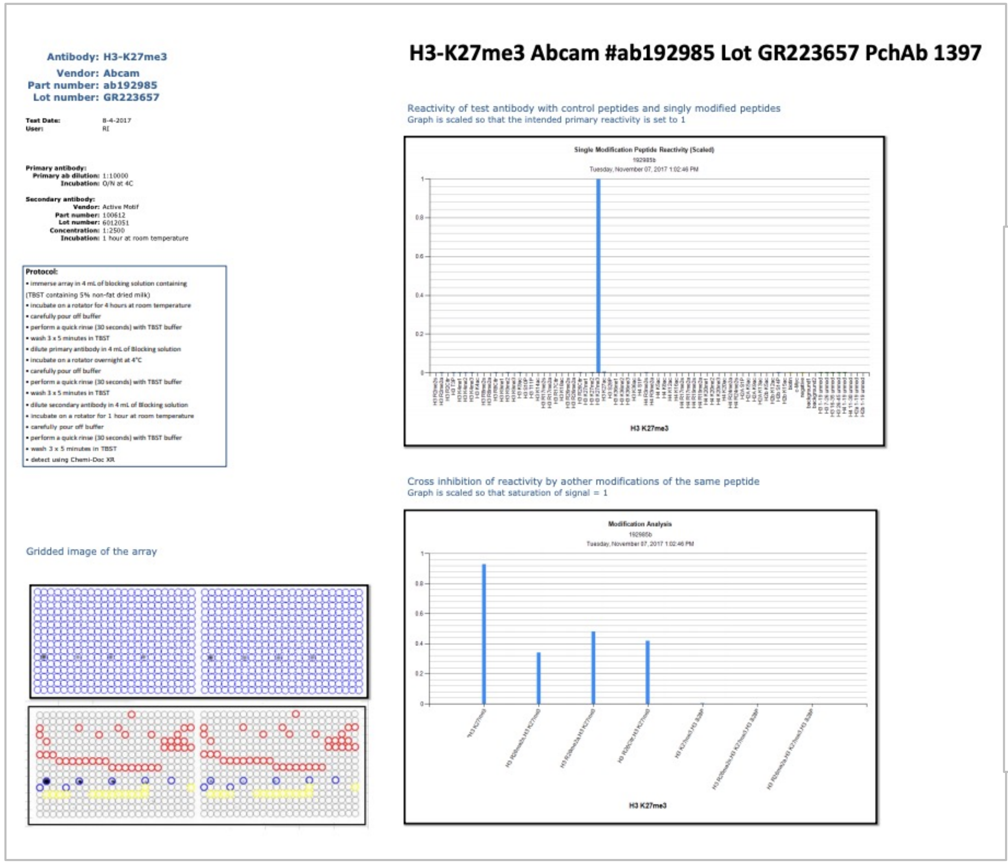
esophagus squamous epithelium

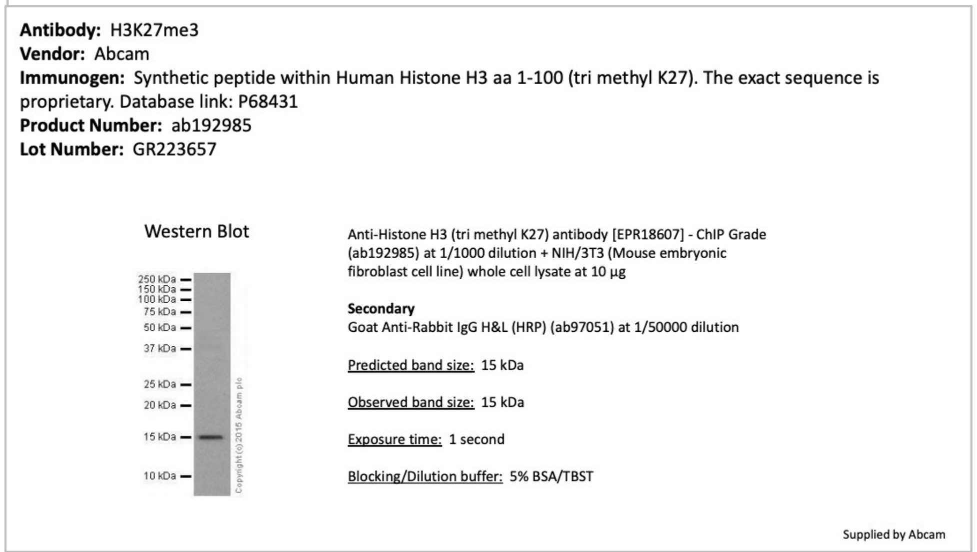
Method: immunoblot

exempt from standards

2. Antibody characterizations

- https://www.encodeproject.org/search/?type=AntibodyLot&status=released




Antibody: H3K27me3
Vendor: Abcam
Immunogen: Synthetic peptide within Human Histone H3 aa 1-100 (tri methyl K27). The exact sequence is proprietary. Database link: P68431
Product Number: ab192985
Lot Number: GR223657

Anti-Histone H3 (tri methyl K27) antibody [EPR18607] - ChIP Grade (ab192985) at 1/1000 dilution + NIH/3T3 (Mouse embryonic fibroblast cell line) whole cell lysate at 10 µg

Secondary
Goat Anti-Rabbit IgG H&L (HRP) (ab97051) at 1/50000 dilution

Predicted band size: 15 kDa

Observed band size: 15 kDa

Exposure time: 1 second

Blocking/Dilution buffer: 5% BSA/TBST

Supplied by Abcam

3. Reference filesets

- https://www.encodeproject.org/data-standards/reference-sequences/

Reference Sequences

Genome References

The ENCODE project uses Reference Genomes from NCBI or UCSC to provide a consistent framework for mapping high-throughput sequencing data. In general, ENCODE data are mapped consistently to 2 human (GRCH38, hg19) and 2 mouse (mm9/mm10) genomes for historical comparability. *Drosophia melanogaster* experiments are mapped to either dm3 or dm6 and *Caenorhabdilis elegans* experiments are mapped to ce10 or ce11. The official reference files for each Uniform processing pipeline can be found in the table below, organized by organism and pipeline. In addition to the genome sequences (we generally use the "no alt" version for each genome), a variety of other crucial files can be found there as well (GENCODE transcript references, chromosome size files, the phage lambda genome, etc.).

Reference File Sets

The table below includes files used by each pipeline for uniform processing by the ENCODE DCC, with associated details on genome assembly and annotation, if applicable. For your convenience, the GRC genome assembly and GENCODE annotation files are directly linked below. For further information, please contact encode-help@lists.stanford.edu

ENCODE4 Uniform Processing Pipeline Filesets

Organism	Pipeline(s)	Reference file set	Genome assembly & annotation
	ATAC-seq	ENCSR938RZZ	GRCh38 with GENCODE version V29
	Bulk RNA-seq	ENCSR151GDH	GRCh38 with GENCODE version V29
	ChIP-seq	ENCSR174MOP (TF, Histone)	GRCh38
		ENCSR487FYI (MINT-ChIP)	GRCh38
	DNase-seq	ENCSR083JOX	GRCh38
	Long read RNA-seq	ENCSR925QOG	GRCh38 with GENCODE version V29
	microRNA-seq	ENCSR608ULQ	GRCh38 with GENCODE version V29

Pipeline search

Clear Filters

Assay names

Status

released 4

Title

Software

Lab

accession

Selected filters: ENCPL612HIG ENCPL809GEM ENCPL367MAS ENCPL481MLO

ENCPL612HIG
ENCPL809GEM
ENCPL367MAS
ENCPL481MLO

Showing 4 of 4 results

Report

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Histone ChIP-seq 2 (unreplicated) — Pipeline ENCPL809GEM released
Assays: ChIP-seq, Mint-ChIP-seq
Software: Bowtie 2, Samtools, Sambamba, Picard, cutadapt, ptools, MACS, BEDTools, overlap_peaks.py, bedToBigBed

Histone ChIP-seq 2 — Pipeline ENCPL612HIG released
Assays: ChIP-seq, Mint-ChIP-seq
Software: Bowtie 2, Samtools, Sambamba, Picard, cutadapt, ptools, MACS, BEDTools, overlap_peaks.py, bedToBigBed

Transcription factor ChIP-seq 2 — Pipeline ENCPL367MAS released
Assays: ChIP-seq, Mint-ChIP-seq
Software: Bowtie 2, Samtools, Sambamba, Picard, cutadapt, ptools, MACS, BEDTools, Irreproducible Discovery Rate (IDR), Phantompeakqualtools, bedToBigBed

Transcription factor ChIP-seq 2 (unreplicated) — Pipeline ENCPL481MLO released
Assays: ChIP-seq, Mint-ChIP-seq
Software: Bowtie 2, Samtools, Sambamba, Picard, cutadapt, ptools, MACS, BEDTools, Irreproducible Discovery Rate (IDR), Phantompeakqualtools, bedToBigBed

3. Reference filesets

- https://www.encodeproject.org/data-standards/reference-sequences/

Reference Sequences

Genome References

The ENCODE project uses Reference Genomes from NCBI or UCSC to provide a consistent framework for mapping high-throughput sequencing data. In general, ENCODE data are mapped consistently to 2 human (GRCH38, hg19) and 2 mouse (mm9/mm10) genomes for historical comparability. *Drosophia melanogaster* experiments are mapped to either dm3 or dm6 and *Caenorhabdilis elegans* experiments are mapped to ce10 or ce11. The official reference files for each Uniform processing pipeline can be found in the table below, organized by organism and pipeline. In addition to the genome sequences (we generally use the "no alt" version for each genome), a variety of other crucial files can be found there as well (GENCODE transcript references, chromosome size files, the phage lambda genome, etc.).

Reference File Sets

The table below includes files used by each pipeline for uniform processing by the ENCODE DCC, with associated details on genome assembly and annotation, if applicable. For your convenience, the GRC genome assembly and GENCODE annotation files are directly linked below. For further information, please contact encode-help@lists.stanford.edu

ENCODE4 Uniform Processing Pipeline Filesets

Organism	Pipeline(s)	Reference file set	Genome assembly & annotation
	ATAC-seq	ENCSR938RZZ	GRCh38 with GENCODE version V29
	Bulk RNA-seq	ENCSR151GDH	GRCh38 with GENCODE version V29
	ChIP-seq	ENCSR174MOP (TF, Histone)	GRCh38
		ENCSR487FYI (MINT-ChIP)	GRCh38
	DNase-seq	ENCSR083JOX	GRCh38
	Long read RNA-seq	ENCSR925QOG	GRCh38 with GENCODE version V29
	microRNA-seq	ENCSR608ULQ	GRCh38 with GENCODE version V29

Summary for reference file set ENCSR938RZZ

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Summary

Status:	released
Accession:	ENCSR938RZZ
Description:	Reference & index files used by ENCODE4 Uniform Processing ATAC-seq pipelines for GRCh38 for version 29
Organism:	human
Related Pipelines:	ENCPL344QWT, ENCPL787FUN

Attribution

ENCODE PHASE 4

Lab:	ENCODE Processing Pipeline
Award:	U24HG009397 (J. Michael Cherry, Stanford)
External resources:	*None submitted*

Files

GRCh38 V29 Visualize

Displaying 9 of 9 files

Processed data

Accession	File type	Output type	Isogenic replicate	Mapping assembly	Lab	Date added	File size	Audit status	File status
ENCFF356LFX	bed bed3	exclusion list regions		GRCh38	Anshul Kundaje, Stanford	2020-05-05	8.02 kB	✓	released

Reference data

Accession	File type	Output type	Mapping assembly	Genome annotation	Lab	Date added	File size	Audit status	File s
ENCFF110MCL	tar	genome index	GRCh38		ENCODE Processing Pipeline	2019-10-17	3.49 GB	✓	

Training

- Factor: CTCF, RNAPII, p53
- Cell line: A549
- Data:

  ChIP-seq

  RNA-seq

  ChIA-PET

  HI-C

  ATAC-seq